

Sample Exam

Medical Image Analysis

Student name: _____

Immatriculation number: _____

Exam Questions

Note: All the questions have equal weight.

1. In 2D acquisitions, 2D image slices are acquired in one pre-defined direction one after the other, and the final 3D volume is created by stacking the 2D slices. The voxel sizes are given as in-plane (two values, the pixel size within the imaging plane) and through-plane (one value, the slice thickness). Recall that the field-of-view is the real-world size of the entire image, i.e. image size $\times$ voxel size. Based on this please compute the following.

- (a) A *coronal* acquisition with voxel size in-plane $0.9 \times 0.9 \text{ mm}^2$ with 260 pixels in both directions and voxel size through-plane 3.0 mm with 64 pixels is acquired. What is the field-of-view in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

Solution: Answer: $234 \times 234 \times 192 \text{ mm}^3$.

The coronal plane contains the head-to-toe and the left-to-right directions, so both in-plane directions give $260 \times 0.9 = 234 \text{ mm}$. The through-plane direction of a coronal acquisition is anterior-to-posterior: $64 \times 3.0 = 192 \text{ mm}$.

- (b) The protocol has to be changed so that the anterior-to-posterior coverage is at least 210 mm, while the slice thickness stays at 3.0 mm. How many slices have to be acquired?

Solution: Answer: 70 slices.

$210/3.0 = 70$. The 64 slices of the original protocol only cover 192 mm.

- (c) You want to resample the volume of part (a) to isotropic voxels of $1.0 \times 1.0 \times 1.0 \text{ mm}^3$. What are the resize factors and the resulting image size? In which of the three directions is the resampling least justified by the information actually contained in the data, and why?

Solution: Answer: resize factors $0.9 \times 0.9 \times 3.0$, resulting image size $234 \times 234 \times 192$ voxels.

The resize factor per direction is the ratio of the old to the new voxel size: $0.9/1.0 = 0.9$ head-to-toe, $0.9/1.0 = 0.9$ left-to-right and $3.0/1.0 = 3.0$ anterior-to-posterior. Applying these: $260 \cdot 0.9 = 234$, $260 \cdot 0.9 = 234$, $64 \cdot 3.0 = 192$. The field-of-view is unchanged, as it must be.

The anterior-to-posterior direction. It is the through-plane direction, and it is being upsampled by a factor of 3: the interpolation invents two new samples between every

pair of acquired slices. No new information is created — the effective resolution in that direction is still that of a 3 mm slice, even though the voxel grid now says 1 mm.

2. Histogram equalization improves the contrast of an image by finding a monotonically increasing intensity mapping $i \mapsto f(i)$ that flattens the histogram. Let $p(i)$ denote the probability density function of the intensities, $P(i) = \int_0^i p(j) dj$ the cumulative distribution function, and i_{max} the maximum intensity. Which of the two mappings below is the one that flattens the histogram? Please circle the correct choice.

$$f(i) = i_{max} p(i) \qquad f(i) = i_{max} P(i) = i_{max} \int_0^i p(j) dj$$

Solution: Answer: Right one — the scaled *cumulative* distribution function.

Changing variables gives $p(f(i)) = p(i) \frac{di}{df(i)} = p(i) \cdot \frac{1}{p(i)} \cdot \frac{1}{i_{max}} = \frac{1}{i_{max}}$, i.e. a uniform (flat) density. Note also that only the CDF is guaranteed to be monotonically increasing, which the mapping has to be in order not to reorder intensities; the PDF is not.

3. When a Markov random field prior is used, segmentation can be written either as a posterior maximization or as an energy minimization. I is the image, c the labelling, $G(x)$ the neighbours of pixel x , and Z a normalization constant.

$$\max_c \log \left[\prod_{x \in \Omega} p(I(x) | c(x)) \right] + \log \left[\frac{1}{Z} \exp\{-E(c)\} \right]$$

$$\min_c \underbrace{\sum_{x \in \Omega} g(I(x) | \theta_{c(x)})}_{\text{unary term}} + \underbrace{\sum_{x \in \Omega} \sum_{y \in G(x)} d(c(x), c(y))}_{\text{pairwise term}}$$

- (a) Please indicate which term corresponds to which term in the two formulations.

Solution: Answer:

$$\log \prod_x p(I(x) | c(x)) \iff \text{the unary term} \quad (\text{likelihood / data fidelity})$$

$$\log \frac{1}{Z} \exp\{-E(c)\} \iff \text{the pairwise term} \quad (\text{prior / neighbourhood consistency})$$

The correspondence requires the data model to be exponential, i.e. $p(I(x) | c(x)) \propto \exp\{-g(I(x) | \theta_{c(x)})\}$, so that taking the logarithm of the product turns it into the unary sum. The pairwise term is exactly $E(c)$, and $\log Z$ does not depend on c and therefore drops out of the maximization.

- (b) Which theorem guarantees that a prior satisfying the Markov property can be written as a Gibbs distribution $\frac{1}{Z} \exp\{-E(c)\}$ with a locally defined energy E ?

Solution: Answer: the Hammersley–Clifford theorem. It states that any probability distribution satisfying a Markovian property is a Gibbs distribution for an appropriate locally defined energy, and vice versa.

4. Please indicate whether the statement is true or false by circling the correct choice in each of the following.
- (a) **TRUE** False A rigid-body transformation in 3D has 6 degrees of freedom: three rotation angles and three translations.
 - (b) **TRUE** False For *isotropic* scaling, $SRx = RSx$, i.e. the order in which rotation and scaling are applied does not matter.
 - (c) True **FALSE** In homogeneous coordinates, a 3D affine transformation is written as a 3×3 matrix.
 - (d) True **FALSE** The determinant of the Jacobian of a transformation is larger than 1 in the regions that the transformation compresses.

Solution: (a) TRUE — three rotations, one about each axis, plus a translation vector in $\mathbb{R}^3$. In 2D the rigid model has 3 degrees of freedom (one angle, two translations).

(b) TRUE — with $S = sI$ the scaling commutes with any matrix. This is no longer the case for *anisotropic* scaling, where $S = \text{diag}(s_1, s_2, s_3)$: there the order matters a great deal, and applying the factors in the wrong order introduces shearing.

(c) FALSE — it is a 4×4 matrix, $\tilde{A} = \begin{bmatrix} A & t \\ \mathbf{0} & 1 \end{bmatrix}$ with $A \in \mathbb{R}^{3 \times 3}$ and $t \in \mathbb{R}^3$. The 3×3 homogeneous matrix is the 2D case.

(d) FALSE — it is the other way around: $\det(J) > 1$ means local *expansion*, $\det(J) < 1$ local compression, and $\det(J) = 1$ no volume change. The divergence $\nabla \cdot T$, the trace of the Jacobian, carries related information.

5. A free-form deformation (FFD) model with cubic B-splines is used for the non-linear registration of two 3D images. The control points are placed on a uniform grid of $10 \times 12 \times 8$ points. How many parameters does this transformation model have, and how does this compare with the number of parameters of a 3D affine transformation?

Solution: Answer: $10 \cdot 12 \cdot 8 = 960$ control points, each carrying a displacement vector in $\mathbb{R}^3$, hence $960 \times 3 = 2880$ parameters. A 3D affine transformation has 12 parameters (3 rotations + 3 translations + 3 scalings + 3 shears).

The FFD model therefore has roughly 240 times more degrees of freedom, which is exactly the point: the affine model is global and cannot capture local deformation, while the FFD model can. Note that the parameter count of the FFD depends only on the number of control points, not on the number of voxels — that is what makes it tractable compared with one displacement vector per voxel.

6. The N3 algorithm corrects the bias field in MRI. It works in the log domain, where the observed image is $j(x) = b(x) + i(x)$ with b the bias field and i the bias-free image, and where the bias

field is assumed to be independent of the image intensities. Each iteration alternates between two estimation problems. Please briefly describe what each of them does, and explain why the algorithm has to iterate with narrow Gaussians instead of correcting the bias in a single step. You do not need to write down the equations.

Solution: Answer: *Estimating the distribution of the bias-free intensities.* Because b and i are independent and add in the log domain, the observed intensity distribution is the convolution $J = I * B$. Since the multiplicative bias is close to 1, its logarithm is close to 0, so B can be approximated by a Gaussian with a small standard deviation and J is a blurred version of I . The distribution I is therefore recovered by *sharpening* J , using a Wiener-style deblurring filter in the Fourier domain.

Estimating the field. Given I and B , the expected bias-free intensity corresponding to an observed intensity j is $E[i | j]$, so the bias estimate is $b_e(j) = j - E[i | j]$. This is a one-dimensional integral and is evaluated numerically. It is computed at anchor points and then smoothed over space with splines or radial basis functions — this is where the prior knowledge that the bias field is spatially smooth enters — and subtracted from j .

Why iterate: B is not actually known, it is only assumed to be Gaussian. Deconvolving with a wide Gaussian is unstable and can produce estimates that are not valid distributions at all. Since a wide Gaussian is a convolution of narrow ones, the correction is instead applied as a sequence of small, stable steps. Iterating also compensates for the other approximations made along the way: the noise term was ignored, and the deblurring filter is itself only approximate.

7. Which of the following statements on the loss metrics used for intensity-based registration is **inaccurate**?

- ☐ (A) The sum of squared differences assumes that the intensities of the two images are directly comparable, which restricts it to intra-modality registration.
- ☐ (B) Normalized cross-correlation attains its maximum magnitude when the two images are related by a linear intensity relationship that is the same at every pixel; both positive and negative correlations indicate alignment, which is why the squared value is used in the loss.
- ☐ (C) Mutual information is maximized when the joint intensity distribution equals the product of the two marginals, that is, when the intensities of the two images are statistically independent.
- ☐ (D) Mutual information can capture more complicated, non-linear relationships between the intensities than normalized cross-correlation, but it is harder to compute.

Solution: Answer: C.

It states exactly the condition under which mutual information is *zero*: $MI = 0$ if and only if $p(I, J) = p(I)p(J)$, which means the two images tell us nothing about each other. Registration *maximizes* MI, because aligned images have a strong statistical relationship and their joint distribution is then as far as possible from the product of the marginals.

8. Which of the following statements about interpretability and explainable AI (XAI) in medical image analysis is **incorrect**?

- ☐ (A) Black-box, or model-agnostic, methods such as LIME require no access to the internals of the model, whereas gradient-based methods such as Grad-CAM are white-box methods.
- ☐ (B) Grad-CAM generalizes CAM: the importance weight of a feature map is the global average of the gradients of the class score with respect to that feature map, which makes it applicable to any CNN, while CAM requires a particular architecture.
- ☐ (C) Because saliency maps show which parts of the image the model actually uses, a model whose saliency lies largely outside the anatomical region of interest is necessarily performing poorly on the task.
- ☐ (D) TCAV explains a prediction in terms of human-defined concepts by measuring the directional derivative of the class score along a concept activation vector, and can be used to spot biases in a data set.

Solution: Answer: C.

High accuracy and misplaced attention are entirely compatible — that is precisely what makes shortcut learning dangerous. In the glioma grading case study, normalizing the intensities over the whole image instead of over the brain area *raised* the AUC from 0.8857 to 0.9841, while the saliency maps showed the model attending outside the brain: the model was exploiting an artefact of the preprocessing. The pneumonia classifier that learned to recognize the clinical site from hospital-specific markers is the same story. Saliency outside the region of interest is a reason to distrust a good score, not evidence of a bad one.

9. Which of the following statements about self-attention, as it was built up in the lecture, is **incorrect**?

- ☐ (A) Parameter-free dot-product attention, $Y = \text{Softmax}(XX^T)X$, has no capacity to learn from data, and every feature within a token plays an equal role in determining the attention coefficients.
- ☐ (B) Introducing a single learnable matrix U and computing $Y = \text{Softmax}(XUU^TX^T)XU$ still yields a symmetric matrix of attention scores, which is why separate key and query projections are introduced.
- ☐ (C) The scaling by $1/\sqrt{D_K}$ is used because the gradients of the softmax become very small for very large or very small inputs; if the entries of Q and K have unit variance, the entries of QK^T have variance D_K .
- ☐ (D) In multi-head self-attention the outputs of the individual heads are averaged, so that the output dimension equals D_V irrespective of the number of heads.

Solution: Answer: D.

The heads are *concatenated*, not averaged, and the concatenation is then projected back: $Y = \text{Concat}(H_1, \dots, H_H)W_0$ with $\text{Concat}(\cdot) \in \mathbb{R}^{N \times HD_V}$ and $W_0 \in \mathbb{R}^{HD_V \times D}$. Averaging would defeat the purpose of using several heads, which — by analogy with the multiple filters of a convolutional layer — is to let different heads attend to different patterns.

10. In the Noise2Noise setting you have no clean images at all. For each input x_n you only have several *noisy* versions y_n^m of the underlying clean image y_n , and you train with $\arg \min_{\theta} \sum_n \sum_m \|y_n^m - f(x_n; \theta)\|^2$. Under which condition does this give the same solution as supervised training on clean targets, in the limit of infinite data?

- ☐ (A) When the noise is multiplicative rather than additive.
- ☐ (B) When the average of the noisy targets is the clean image, $\mathbb{E}[\hat{y} \mid x] = y$, because the prediction that minimizes an ℓ_2 loss is the mean of the targets.
- ☐ (C) When at least half of the training targets are clean images.
- ☐ (D) When an adversarial loss is added to the ℓ_2 loss, so that the outputs are forced to look realistic.
- ☐ (E) Never — training on noisy targets always biases the solution towards the noise.

Solution: Answer: B.

Minimizing $\mathbb{E}_{\hat{y}}[\|f(x; \theta) - \hat{y}\|^2]$ over the prediction gives $f(x; \theta) = \mathbb{E}_{\hat{y}}[\hat{y}]$: the ℓ_2 -optimal prediction is the mean of the targets it is trained against. So if the noisy targets are distributed around the clean image, $\mathbb{E}[\hat{y} \mid x] = y$, the network converges to the clean image even though it never sees one. The network cannot fit the individual noise realizations, because a single set of parameters has to satisfy all of them at once, and only their mean does.

11. Consider a classification network that takes grey-scale input images of size 64×64 with one input channel. All convolutions are *valid* (no padding, $P = 0$), and the network uses strided convolutions instead of pooling:

- **Conv1:** 16 kernels with kernel size $K = 5 \times 5$ and stride $S = 2$ in each direction.
- **Conv2:** 32 kernels with $K = 5 \times 5$ and $S = 2$.
- **Conv3:** 64 kernels with $K = 3 \times 3$ and $S = 2$.

Recall the channel size of a strided, valid convolution: $N_l = \lceil (N_{l-1} - k_1)/s_1 \rceil$. Circle True or False:

- (a) **TRUE** False The output of Conv3 has a spatial size of 5×5 .
- (b) True **FALSE** The global receptive field of a neuron in Conv3 is 13×13 .
- (c) True **FALSE** The global stride of a neuron in Conv3 is 6×6 , since the three strides of 2 add up.
- (d) **TRUE** False The total number of convolutional kernel weights (excluding the biases) is $5 \cdot 5 \cdot 1 \cdot 16 + 5 \cdot 5 \cdot 16 \cdot 32 + 3 \cdot 3 \cdot 32 \cdot 64 = 31\,632$.

Solution: (a) TRUE — applying $N_l = \lceil (N_{l-1} - k)/s \rceil$ layer by layer: Conv1 gives $\lceil (64 - 5)/2 \rceil = \lceil 29.5 \rceil = 30$; Conv2 gives $\lceil (30 - 5)/2 \rceil = \lceil 12.5 \rceil = 13$; Conv3 gives $\lceil (13 - 3)/2 \rceil = 5$.

(b) FALSE — it is 21×21 . Tracking the receptive field R and the jump j from $R = 1, j = 1$ with $R \leftarrow R + (K - 1) \cdot j$ and $j \leftarrow j \cdot S$: Conv1 $\rightarrow R = 5, j = 2$; Conv2 $\rightarrow R = 13, j = 4$; Conv3 $\rightarrow R = 21, j = 8$. The value 13 is the receptive field one layer earlier, after Conv2.

(c) FALSE — strides compound multiplicatively: $2 \cdot 2 \cdot 2 = 8$, so the global stride is 8×8 .

(d) TRUE — $400 + 12\,800 + 18\,432 = 31\,632$. Note that strided convolutions reduce the spatial dimensions quickly, but unlike max pooling they do *not* buy any translation invariance, and they discard information — the more so the larger the stride.

12. Which of the following statements on restoration, synthesis and self-supervised learning are correct (multiple answers are possible)?

- ☐ (A) Losses of the MSE type tend to produce over-smoothed restorations, which is one motivation for perceptual losses, where the two images are passed through a separately trained network and their intermediate features are compared.
- ☐ (B) In CycleGAN-style synthesis from unpaired data, the cycle-consistency term requires that mapping a source image to the target domain and back returns the original image.
- ☐ (C) In contrastive learning, the two views of an image have to be produced by geometric transformations only, since intensity changes would destroy the surrogate label.
- ☐ (D) A masked autoencoder uses a larger encoder on the visible parts of the image and a smaller decoder to reconstruct the masked parts; the trained encoder is then used as a backbone for downstream tasks.
- ☐ (E) Adversarial losses compare individual pairs of images, which is what makes them suitable for unpaired data sets.

Solution: Answer: (A), (B) and (D) are correct.

(C) is wrong: the transformations used in contrastive learning are explicitly *not* restricted to geometric ones and routinely include intensity changes — the whole point is that the representation should be invariant to them. (The one place where transformations do have to be handled carefully is the *local* contrastive loss, where they must not destroy the patch correspondences, so they are either restricted or undone before the patches are compared.)

(E) is wrong in its premise, and the premise is what makes the conclusion work. An adversarial loss is a *distributional* distance: the discriminator is not shown a pair of corresponding images, it is asked whether a single image looks like it came from the real set. That is exactly why it can be used when no correspondence between source and target images exists.